AI assisted Coding lab (4.3)

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Task 1

Prompt and Code

```
# Generate a Python function that accepts a year as input.
# Check whether the given year is a leap year.
# Return an appropriate result.
# Do not use input-output examples.
# Also provide sample test cases with their outputs.
def check_leap_year(year):
    if year % 400 == 0 or (year % 4 == 0 and year % 100 !=
0):
        return "Leap year"
    else:
        return "Not a leap year"


print(check_leap_year(2024))
print(check_leap_year(2023))
```

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ C:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDrive/Desktop/c_prog/Github/4.3.py
Leap year
Not a leap year
```

Output

```
Leap year
```

```
Not a leap year
```

Task 2

Prompt and Code

```
# Generate a Python function that converts centimeters to i
nches.
# Use the correct mathematical conversion formula.
# The function should accept centimeters as input and retur
n the value in inches.
# Example:
# Input: 10 cm
# Output: 3.94 inches
# Also provide additional sample test cases with outputs.
def cm_to_inches(cm):
    return round(cm / 2.54, 2)


print(cm_to_inches(10))
print(cm_to_inches(20))
print(cm_to_inches(2.54))
```



Output

```
3.94
7.87
1.0
```

Task 3

Prompt and Code

```
# Generate a Python function that accepts a full name and f
ormats it as "Last, First".

# Examples:
# "John Smith" -> "Smith, John"
# "Anita Rao" -> "Rao, Anita"
# "Rahul Kumar" -> "Kumar, Rahul"

# Follow the same output pattern for other names.
# Also provide sample inputs and outputs.
def format_name(full_name):
    parts = full_name.split()

    if len(parts) < 2:
        return full_name

    first_name = parts[0]
    last_name = " ".join(parts[1:])

    return f"{last_name}, {first_name}"


print(format_name("John Smith"))
print(format_name("Anita Rao"))
print(format_name("Rahul Kumar"))
```

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ C:/Users/tanik/.local/bin/python3.13.exe "c:/Users/tanik/OneDrive/Desktop/c_prog/Github/4.3/task 3.py"
Smith, John
Rao, Anita
Kumar, Rahul
```

Output

```
Smith, John
Rao, Anita
Kumar, Rahul
```

Task 4

Prompt and Code(Zero Shot)

```python
# Generate a Python function that counts the number of vowe
ls in a given string.
# The function should accept a string as input and return t
he total number of vowels.
# Consider both uppercase and lowercase vowels.
# Also provide sample test cases and outputs.
def count_vowels_zero_shot(text):
    count = 0

    for char in text:
        if char.lower() in "aeiou":
            count += 1

    return count


print(count_vowels_zero_shot("Hello World"))
print(count_vowels_zero_shot("Python Programming"))
```

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ C:/Users/tanik/.local/bin/python3.13.exe "c:/Users/tanik/OneDrive/Desktop/c_prog/Github/4.3/task 4.py"
3
4
```

Output

```
3
4
```

Prompt and Code(Few Shot)

```
# Generate a Python function that counts the number of vowe
ls in a string.

# Example 1:
# Input: "Hello"
# Output: 2

# Example 2:
# Input: "Python"
# Output: 1

# Example 3:
# Input: "Education"
# Output: 5

# The function should handle both uppercase and lowercase v
owels.
# Return only the total number of vowels.
def count_vowels_few_shot(text):
    vowels = "aeiouAEIOU"
    return sum(1 for char in text if char in vowels)


print(count_vowels_few_shot("Hello"))
print(count_vowels_few_shot("Python"))
print(count_vowels_few_shot("Education"))
```

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ C:/Users/tanik/.local/bin/python3.13.exe "c:/Users/tanik/OneDrive/Desktop/c_prog/Github/4.3/task 4.py"
2
1
5
```

Output (Few Shot)

```
2
1
```

```
5
```

Task 5

Prompt and Code

```
# Generate a Python function that reads a .txt file and cou
nts the number of lines.

# Example 1:
# File content:
# Hello
# World
# Output: 2 lines

# Example 2:
# File content:
# Python
# Programming
# Language
# Output: 3 lines

# The function should accept a file path as input.
# Read the file and return the total number of lines.
# Also provide a sample input file and explain the logic.
import os


def count_lines(file_path):
    with open(file_path, "r") as file:
        lines = file.readlines()

    return len(lines)


# Get the folder where task 5.py is located
```

```python
current_folder = os.path.dirname(os.path.abspath(__file__))

# Find sample.txt in the same folder as task 5.py
file_path = os.path.join(current_folder, "sample.txt")

print("Number of lines:", count_lines(file_path))
```

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ C:/Users/tanik/.local/bin/python3.13.exe "c:/Users/tanik/OneDrive/Desktop/c_prog/Github/4.3/task 5.py"
Number of lines: 3
```

Output

```
Number of lines: 3
```